Capstone Project Sprint 2

FixFlow

Maintenance Management System

Agenda

Project Description

Descriptive Minimal Viable Product

Diagrams

Acceptance Criteria & User Strories

Sprint backlog and Test case summary

Team Metrics

Team Members

Viraj Chokshi

Developer and DBA

Shriken Patel

Developer

Harshil Goti

Developer

Krish Jasani

Developer

Improvements made from professor's feedback

- User Stories & acceptance criteria: Improved from the previous presentation, according to the template provided.
- Slides Sequence: The slides are following the checklist sequence unlike the previous presentation.

Project Description

FixFlow is a mobile-first maintenance management ecosystem designed to eliminate the inefficiencies of traditional facility reporting through a photo-centric, "smart triage" approach. By mandating visual evidence during the submission process and utilizing a backend priority engine, the platform automatically categorizes and ranks work orders based on severity, ensuring that critical issues like leaks or electrical hazards are addressed immediately. The system bridges the gap between residents and management by providing tenants with live status tracking and administrators with a data-rich dashboard, backed by a robust audit log that ensures 100% transparency and accountability for every repair.

Team Working Agreement

TEAM AGREEMENT

PACE UNIVERSITY

Introduction

The purpose of this team working agreement is to outline standardized expectations for the Lightning Bolt project concerning, but not limited to, the working relations and group structure among team members in CS-691. The contents herein addressed are:

1. Communication
2. Decision making
3. Responsibility
4. Participation
5. Leadership
6. Consequences

The members of the team are:

1. Harshil Hiteshbhai Goti
2. Krish Kanubhai Jasani
3. Shriken Jigar Patel
4. Viraj Rajeshkumar Chokshi

Communication:

Communication between team members shall be through e-mail, phone conversations, and weekly team meetings. Members will check their e-mail once daily and reply when requested or necessary. Team meetings are scheduled every Tuesday evening at 6:00 pm. If a member cannot attend a team meeting, they must communicate to all members 24 hours prior to the meeting. Failure to communicate their absence will result in a strike (see "Consequences" below).

Decision Making

All ideas and directions will be kept open until a final consensus decision is made by the group. Final ideas and decisions will be adopted in one of two ways: secret ballot or coin toss. Failure to communicate or acting on a decision not sanctioned by the entire team, will result in one strike. (Please refer to "Consequences" section.)

Responsibility

Members of the team are expected to complete all tasks assigned to them by the due date. If unforeseen obstacles prevent task completion, this will be handled accordingly. Difficult or unclear responsibilities must be voiced to other team members swiftly so that they can be clarified or redefined.

Leadership

Leadership is strictly informal with a democratic debate system used for decision-making. A primary meeting facilitator will be assigned prior to each meeting. The facilitator will be responsible for compiling an agenda and directing the smooth flow of the meeting. Natural leadership will evolve over time, and this working agreement shall be edited to accommodate such future logistical changes.

Group Progress

The group will create a timeline that includes dates for expected completion of work and other group objectives. This timeline will help the group to determine progress and how rules should be enforced regarding participation of each group member.

Consequences

Consequences will be based on a strike program with three strikes resulting in a probationary status and four strikes resulting in removal from the team. While on probation the team member must demonstrate his/her ongoing commitment to the team by writing an explanatory paper and requesting reinstatement to the team.

Strikes may be given for any one of the following reasons:

1. Missed meetings without either communication 24 hours prior or a legitimate conflict.
2. Failure to abide by the rules presented in this working agreement.
3. Low commitment and substandard work presented in assigned tasks.

Summary

The ideas and requirements set forth in this working agreement are established to provide the best possible working conditions for completing the assigned project.

If you have any questions concerning this memo, or the contents therein, please direct them to the group secretary:

Viraj Chokshi,

vc62862n@pace.edu

(718)4139344.

Persona 1

Profile

Marcus is a 20-year-old Junior at Pace University living in an on-campus residence hall. He is a full-time student with a heavy course load and a part-time job. He doesn't have time to wait on hold with campus security or visit the facilities office in person. He relies entirely on his smartphone to manage his life.

•Name: Marcus
•Age: 20
•Location: New York, NY (Pace University Housing)
•Status: Student / Resident
•Tech Literacy: High (Prefers apps over phone calls)
Living Situation: Shared Suite

Frustrations:
•Not knowing if his maintenance request was actually received.
•Maintenance staff showing up when he is in the middle of a remote exam.
•Having to explain the same issue multiple times to different people.
Goals:
•Reporting a problem in under 30 seconds.
•Getting real-time notifications about when a technician will arrive.
•Having a functional, safe living space to study in.

Persona 2

Profile

Marcus is a 20-year-old Junior at Pace University living in an on-campus residence hall. He is a full-time student with a heavy course load and a part-time job. He doesn't have time to wait on hold with campus security or visit the facilities office in person. He relies entirely on his smartphone to manage his life.Profile: David is a 45-year-old lead maintenance technician with 15 years of experience. He is skilled at his job but often frustrated by the lack of information in work orders. He hates arriving at a "clogged sink" only to realize he needs a specific industrial snake tool that he left back at the central workshop, 15 minutes away.

- **Name: David**
- **Age: 45**
- **Location: Westchester, NY**
- **Job: Facilities Technician**
- **Work Style: Practical, hands-on, mobile.**
- **Family: Married with two kids.**

Frustrations:

- Walking across a large campus only to find he brought the wrong tools for the job.
- Illegible handwritten notes or vague descriptions like "it's broken."
- Getting blamed for delays when he wasn't notified of a priority emergency.

Goals:

- Seeing a photo of the issue before he leaves the shop to prepare.
- Closing out tickets digitally so he doesn't have to do paperwork at the end of the day.
- Efficiently completing his daily route to reduce overtime.

Persona 3

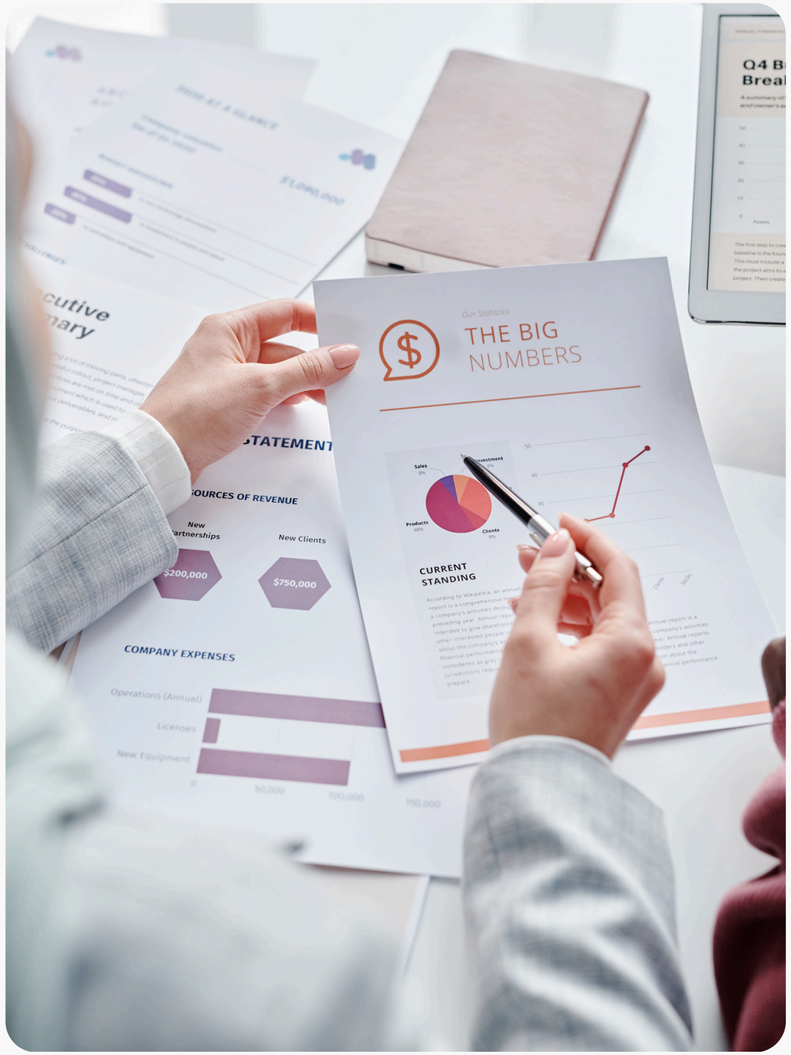
Profile

Profile: Elena is a 38-year-old Property Manager responsible for three large dormitory buildings. She needs to keep costs low and resident satisfaction high. She cares about "the big picture"—knowing which buildings have the most issues and ensuring her team is working efficiently. She is the person who will be interested in the "Audit Logs" you mentioned.

•Name: Elena
•Age: 38
•Location: New York, NY
•Job: Senior Property Manager
•Tech Literacy: Moderate (Wants clean, easy-to-read data)
•Interests: Data analytics, budget optimization.

Frustrations:
•Having no clear data on how long it takes for a ticket to go from "Open" to "Closed."
•Dealing with angry parents/residents complaining about ignored requests.
•Unpredictable repair costs caused by small issues turning into major floods.
Goals:
•Identifying recurring issues (e.g., "Building B has constant pipe leaks") to justify renovations.
•Ensuring the maintenance team is meeting their Service Level Agreements (SLAs).
•Maintaining a 100% digital audit trail for liability and safety compliance.

Minimal Viable Product



Ticket Management

- Priority & category classification
- Assign technician via modal
- Status transitions with audit log
- Attachments + threaded comments

Role Dashboards

- Admin: KPIs, ticket charts, revenue
- Technician: queue & workload
- Finance: revenue & payment stats
- Customer: personal ticket feed

Estimates & Invoices

- Line-item estimates linked to tickets
- One-click convert to invoice
- PDF-ready detail pages
- Status workflow: draft → approved

Technologies

Frontend

React 18
UI Library

TypeScript
Type Safety

Tailwind CSS
Styling

Vite
Build Tool

Backend

Node.js
Runtime

Express
HTTP Framework

MongoDB
Database

Mongoose
ODM

DevOps & Tools

JWT
Authentication

Vercel
Deployment

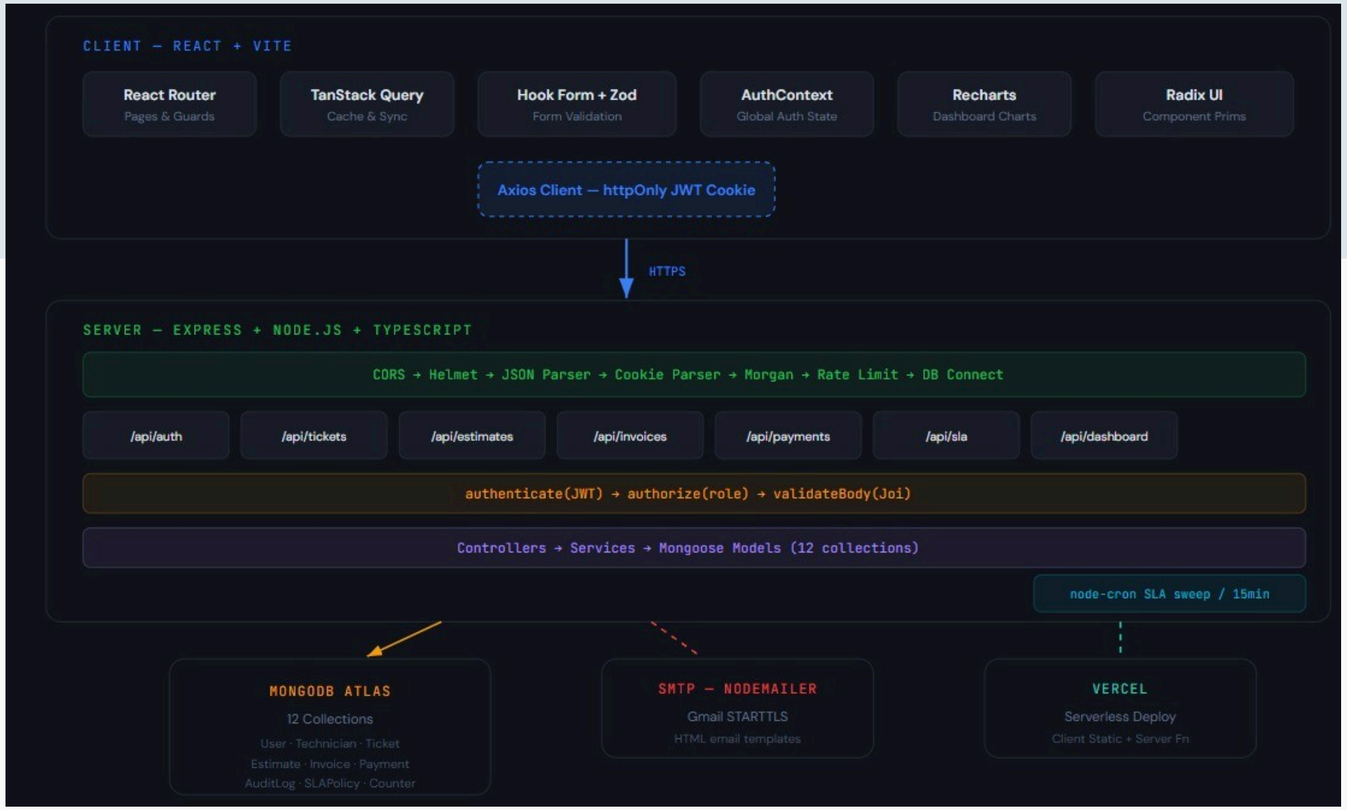
Nodemailer
Email

node-cron
Scheduled Jobs

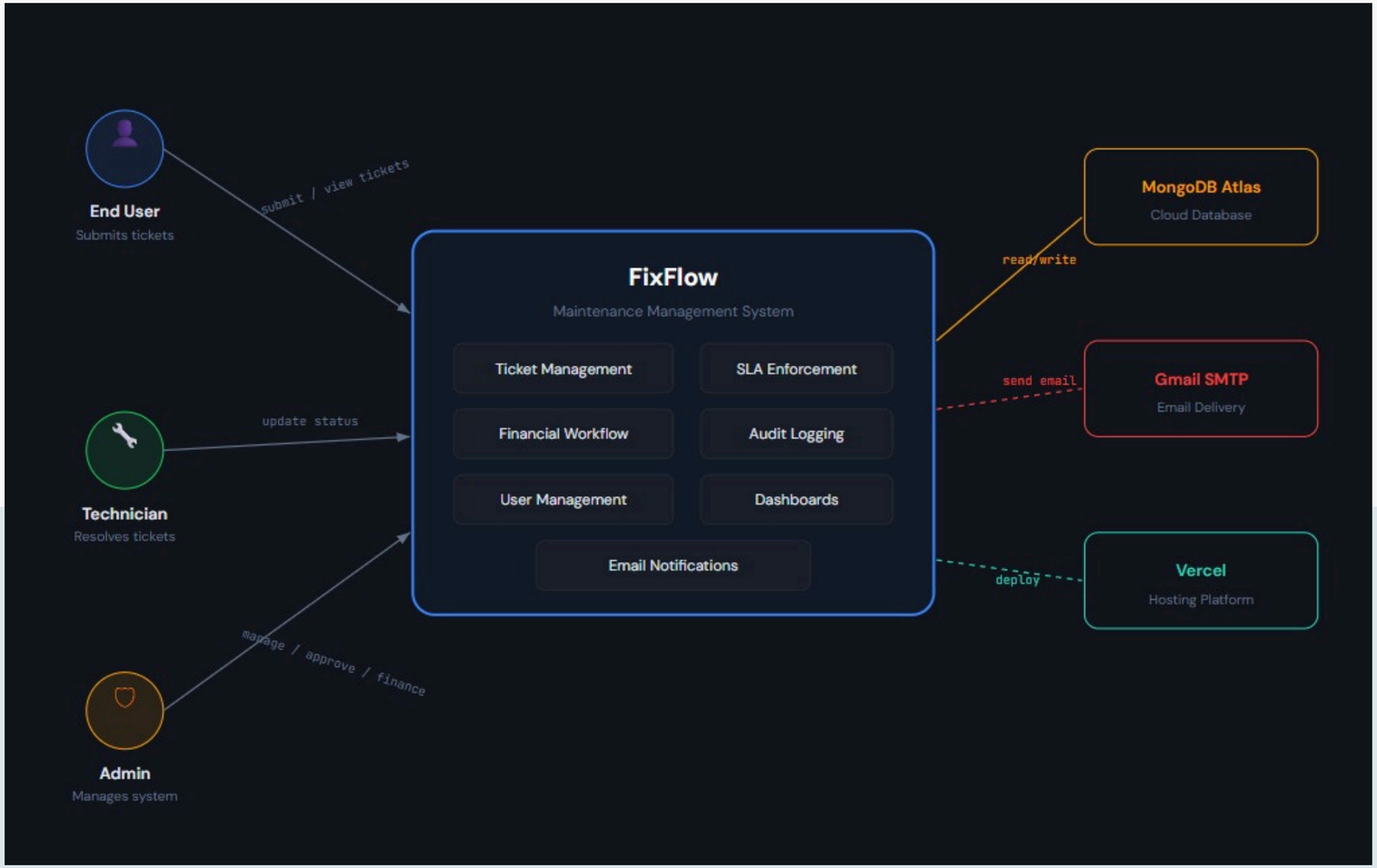
Diagrams

- Architecture Diagram
- Context Diagram
- ER Diagram
- Sequence Diagram
- State Diagram
- Class Diagram

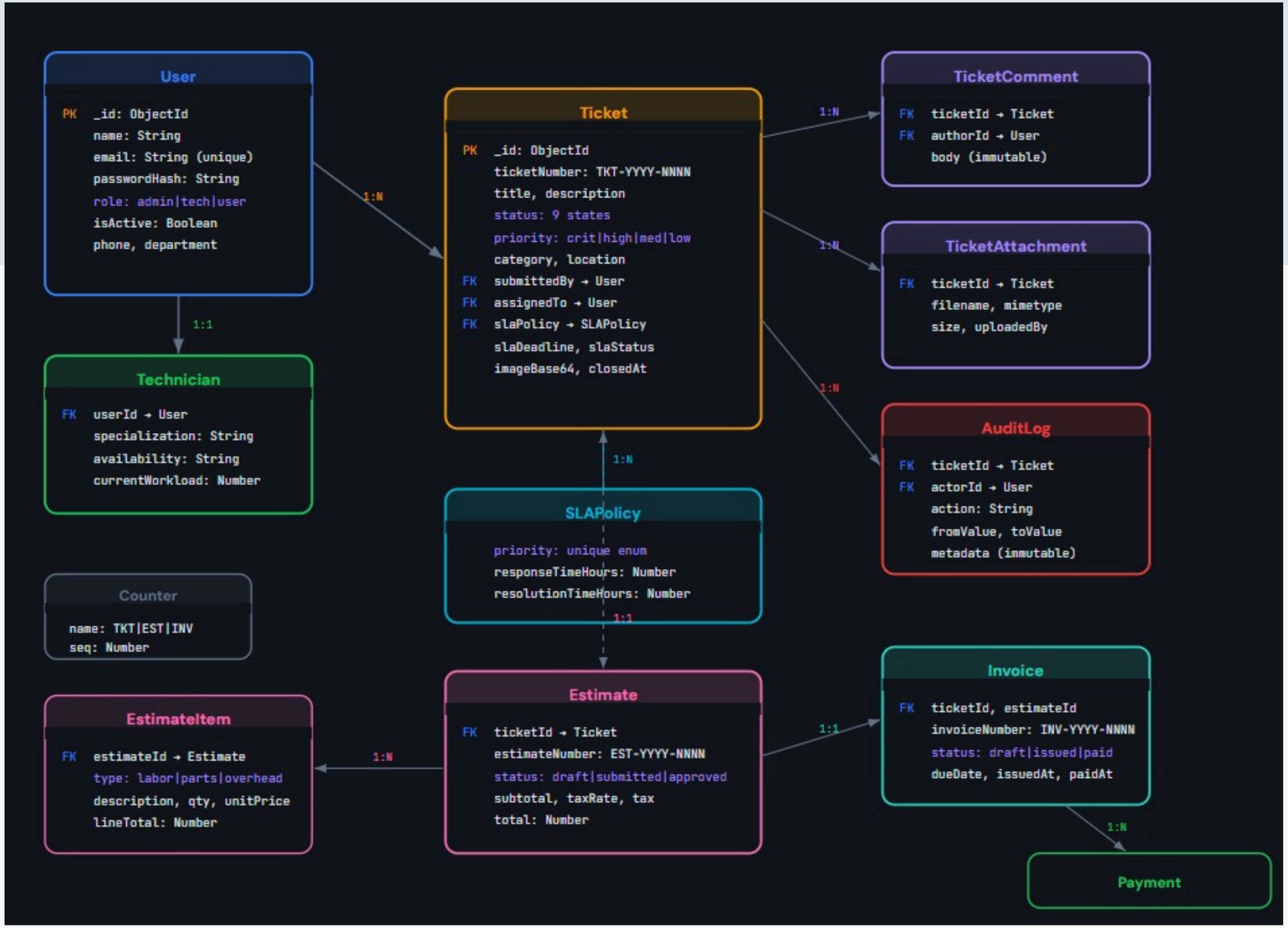
Architecture Diagram



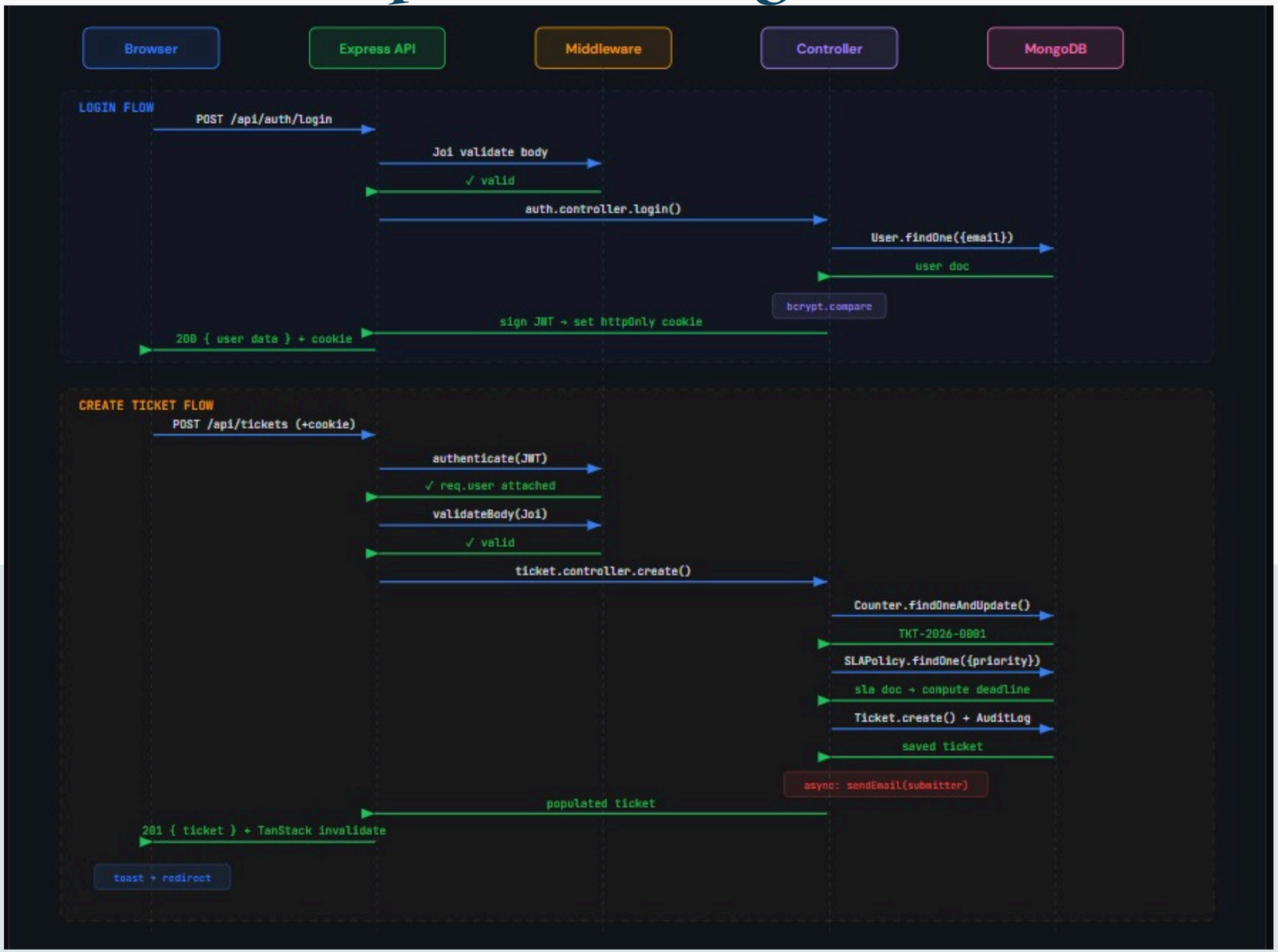
Context Diagram



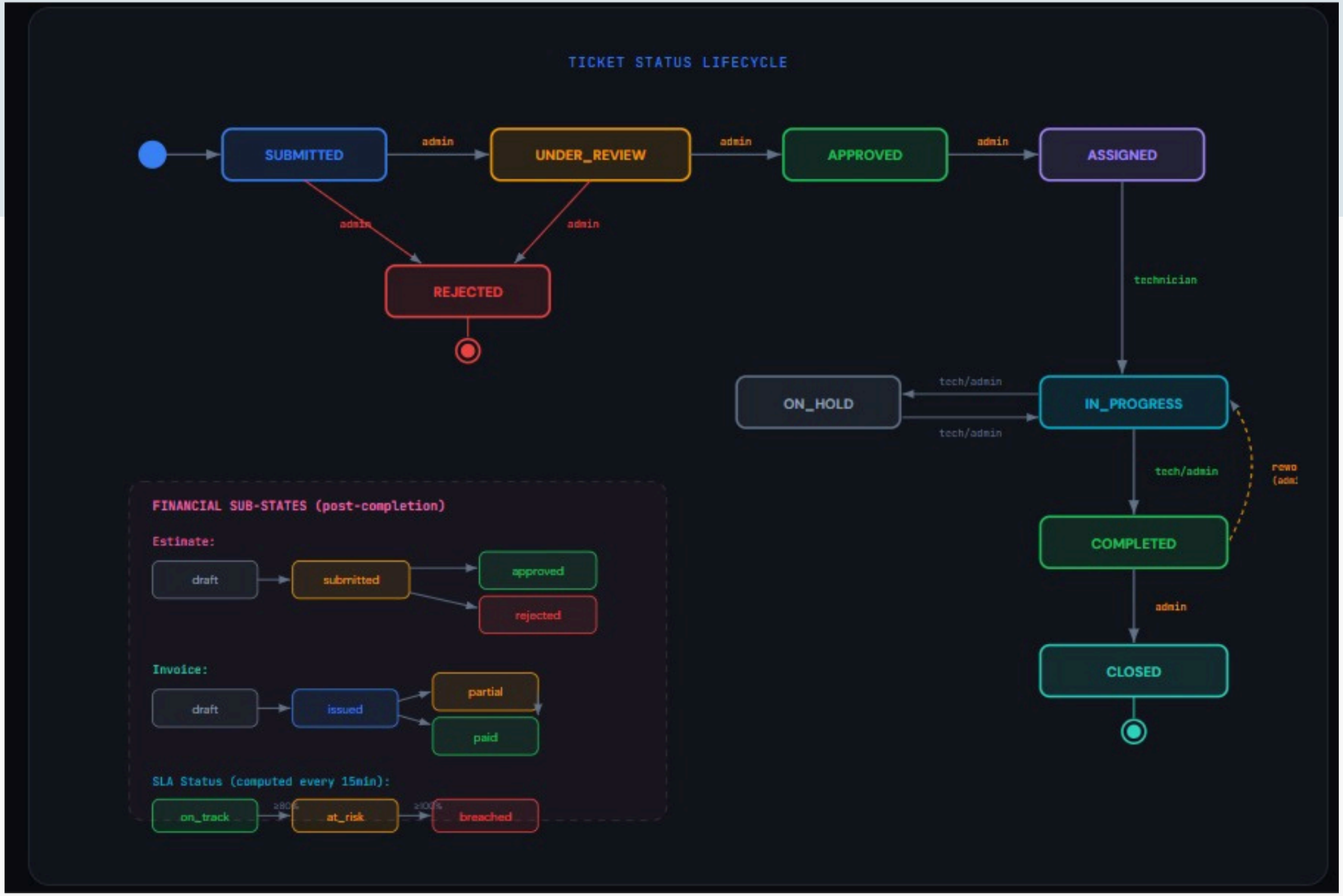
ER Diagram


User
PK _id: ObjectId
name: String
email: String (unique)
passwordHash: String
role: admin|tech|user
isActive: Boolean
phone, department
1:N
1:1
Technician
FK userId → User
specialization: String
availability: String
currentWorkload: Number
Counter
name: TKT|EST|INV
seq: Number
Ticket
PK _id: ObjectId
ticketNumber: TKT-YYYY-NNNN
title, description
status: 9 states
priority: crit|high|med|low
category, location
FK submittedBy → User
FK assignedTo → User
FK slaPolicy → SLAPolicy
slaDeadline, slaStatus
imageBase64, closedAt
1:N
SLAPolicy
priority: unique enum
responseTimeHours: Number
resolutionTimeHours: Number
1:1
Estimate
FK ticketId → Ticket
estimateNumber: EST-YYYY-NNNN
status: draft|submitted|approved
subtotal, taxRate, tax
total: Number
1:N
EstimateItem
FK estimateId → Estimate
type: labor|parts|overhead
description, qty, unitPrice
lineTotal: Number
1:1
Invoice
FK ticketId, estimateId
invoiceNumber: INV-YYYY-NNNN
status: draft|issued|paid
dueDate, issuedAt, paidAt
1:N
Payment
TicketComment
FK ticketId → Ticket
FK authorId → User
body (immutable)
TicketAttachment
FK ticketId → Ticket
filename, mimetype
size, uploadedBy
AuditLog
FK ticketId → Ticket
FK actorId → User
action: String
fromValue, toValue
metadata (immutable)

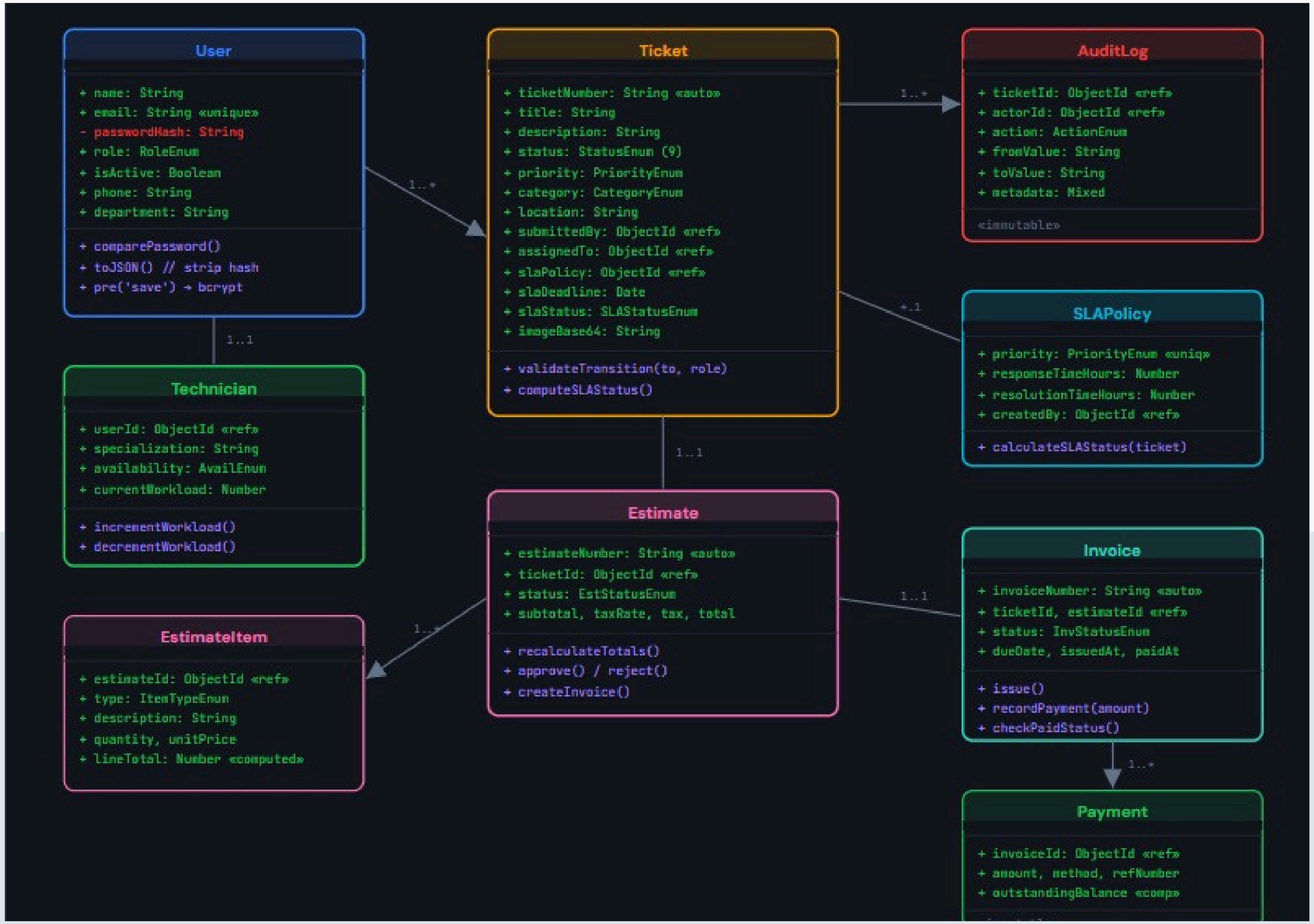
Sequence Diagram



State Diagram



Class Diagram



Acceptance Criteria & User Stories

Authentication

ID	User Story (Role > Action > Benefit)	Acceptance Criteria (Given > When > Then)
US-001	**As a** new user, **I want** to register with my details, **so that** I can gain access to the platform.	**AC-1: Given** a new email, **When** I submit the form, **Then** an account is created with the 'user' role . **AC-2: Given** an existing email, **When** I register, **Then** a 409 Conflict error is returned.
US-002	**As a** registered user, **I want** to log in, **so that** I can access my role-based features.	**AC-1: Given** active credentials, **When** I log in, **Then** a JWT is issued and I am sent to my dashboard . **AC-3: Given** a deactivated account, **When** I log in, **Then** access is rejected.

Ticket Management

ID	User Story (Role > Action > Benefit)	Acceptance Criteria (Given > When > Then)
US-004	**As a** user, **I want** to submit a maintenance request, **so that** the team can begin the repair.	**AC-1: Given** a new ticket form, **When** I submit details, **Then** a TKT-YYYY-NNNN number is assigned . **AC-2: Given** a valid file (<10MB), **When** I upload it, **Then** it is linked to the ticket.
US-005	**As a** user, **I want** to view my tickets, **so that** I can monitor progress.	**AC-1: Given** I am a user, **When** I view the list, **Then** I see only my submitted tickets . **AC-3: Given** a status update, **When** I view the ticket, **Then** the change is reflected in real-time.

SLA Management

ID	User Story (Role > Action > Benefit)	Acceptance Criteria (Given > When > Then)
US-015	**As a** system, **I want** to evaluate SLA status automatically, **so that** admins are alerted to breaches.	**AC-1: Given** an open ticket, **When** the cron runs, **Then** status is set to 'on_track', 'at_risk', or 'breached' . **AC-2: Given** a 'breached' status, **When** sweep ends, **Then** a system-level audit log is created.

Email Notifications

ID	User Story (Role > Action > Benefit)	Acceptance Criteria (Given > When > Then)
US-023	**As a** submitter, **I want** email alerts, **so that** I stay informed without logging in.	**AC-1: Given** a new ticket, **When** created, **Then** a confirmation HTML email is sent . **AC-4: Given** an SMTP failure, **When** a trigger occurs, **Then** the ticket action still completes.

FixFlow — Test Cases Summary

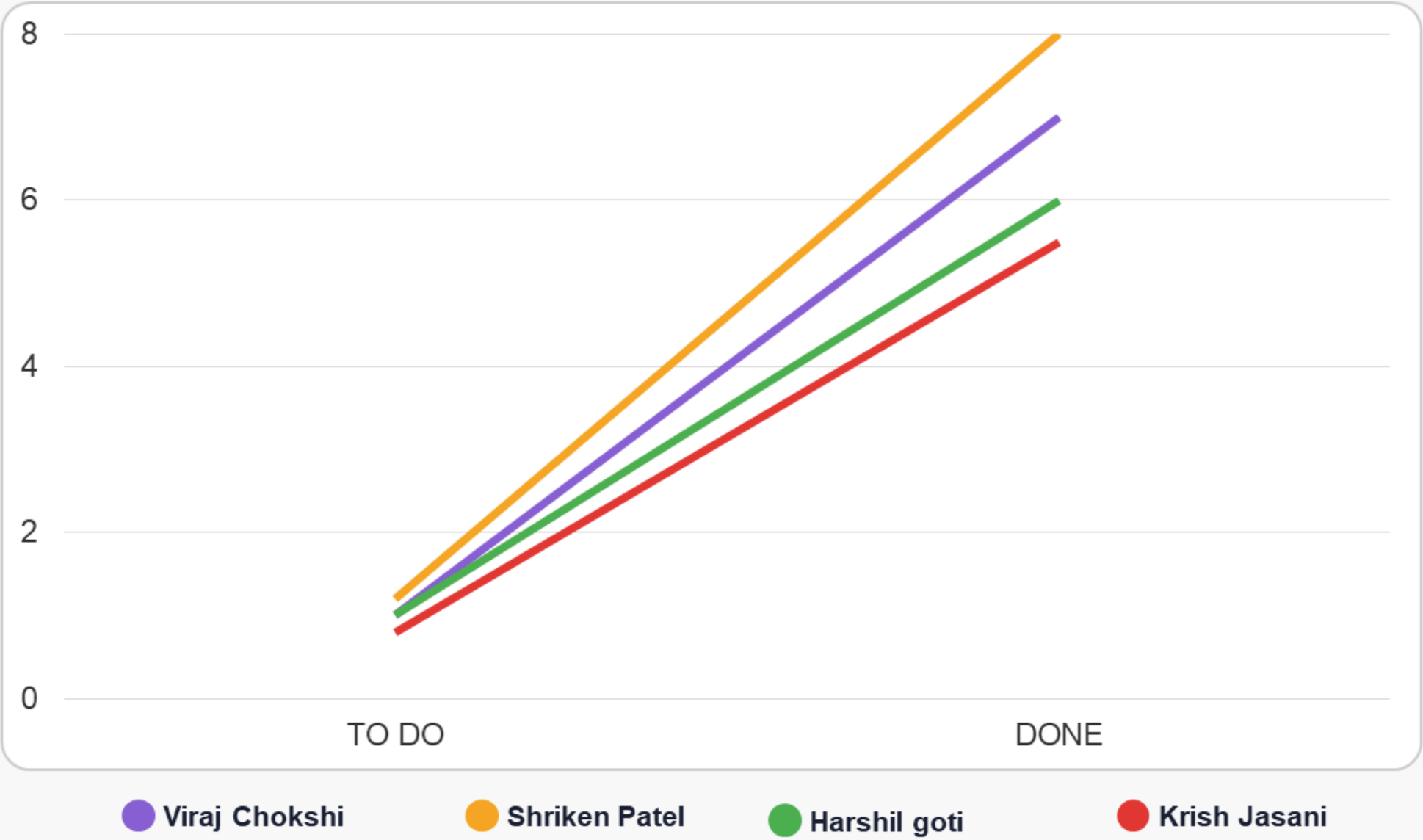
Full-stack Maintenance Management System

Module	Test IDs	Total TCs	Positive	Negative	Status
Authentication	AUTH-001 → AUTH-010	10	4	6	**Completed**
Tickets	TKT-001 → TKT-019	19	10	9	**Completed**
SLA	SLA-001 → SLA-008	8	6	2	**Completed**
Finance	FIN-001 → FIN-012	12	8	4	**Pending**
User Management	USR-001 → USR-006	6	5	1	**Completed**
Dashboard	DSH-001 → DSH-005	5	3	2	**Completed**
Security	SEC-001 → SEC-008	8	4	4	**Pending**
TOTAL		**68**	**40**	**28**	

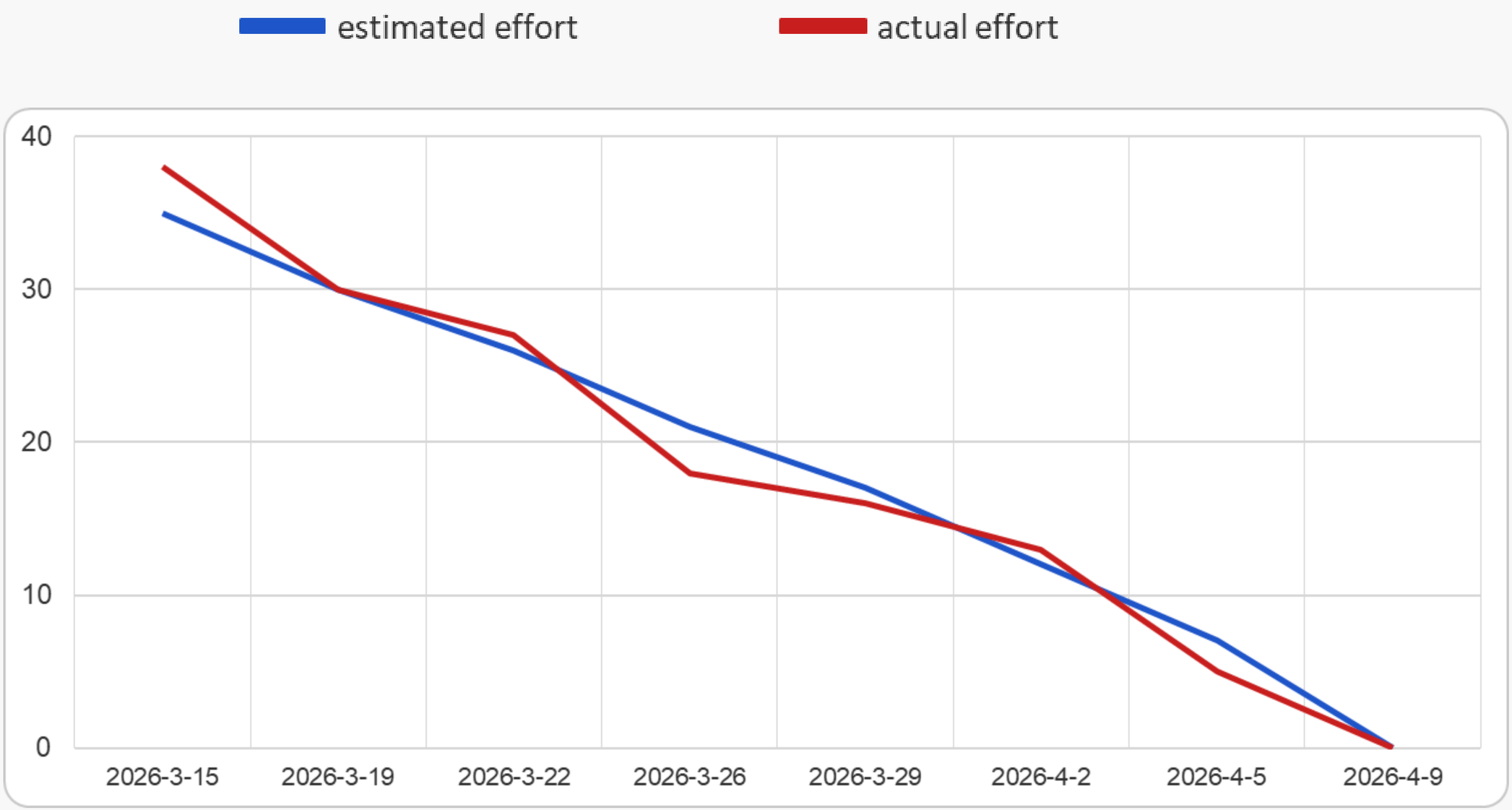
Stories Completed:

- *End-to-End Ticket Lifecycle*
  (Submission → Review → Approval → Assignment → Completion → Closure)
- *SLA Tracking System*
  (Priority-based deadlines with auto monitoring & breach detection)
- *Financial Workflow*
  (Estimates, invoices, and payment tracking per ticket)
- *Email Notifications*
  (Automated updates on ticket status changes)
- *Dashboard & Reporting*
  (KPIs, workload tracking, and system insights)
- *Logging*
  (Full history of all ticket actions)
- *Role-Based Access Control*
  (Admin, Technician, User with secure permissions)

Team members velocity report



Sprint 2 Burndown chart



Sprint Retrospective

Went Well

- Strong team collaboration throughout the sprint
- All critical bugs resolved before demo
- Daily standups stayed focused & time-boxed
- Code reviews completed within 24 hours
- SLA engine delivered on schedule
- Good test coverage on ticket module

Needs Improvement

- Sprint planning took too long (3h+)
- Some stories were under-estimated
- Documentation updated late in sprint
- QA handoff had gaps in acceptance criteria
- Finance module delayed mid-sprint
- Tech debt not being tracked formally

Sprint Retrospective

Stop Doing

- Skipping retrospective action follow-ups
- Direct commits to main branch
- Leaving PR reviews open >48h
- Inconsistent commit message formats
- Unplanned scope additions mid-sprint

Action Items

- Add story point cap: max 3 per unestimated task
- Create DoD checklist for all tickets
- Set up automated linting for commit messages
- Block main branch — PRs required
- Tech debt backlog item added each sprint
- Acceptance criteria mandatory before sprint start

Stories Planned for Sprint 4 (product backlog):

Continue UI/UX enhancements for better usability

Implement advanced reporting dashboards

Improve notification system reliability

Optimize system performance & database queries

Enhance security mechanisms

Set up CI/CD pipeline & monitoring tools

Home

Contribute to Virajchokshi/FixFlow development by creating an account on GitHub.

GitHub

Thank you